Dual-Branch Melanoma Classification with Metadata Fusion, Focal Loss, and Mahalanobis OOD Detection

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***Abstract*—This work presents a dual-branch deep learning system fusing dermoscopic images with patient metadata (age, sex, anatomical site) for binary melanoma classification on the SIIM-ISIC 2020 dataset (13.6% malignant). Using an EfficientNet-B0 backbone with sigmoid focal loss (α=0.864), a hyperparameter search across five experiments (Baseline, Experiments 1–4) isolates the effect of learning-rate scheduling, optimiser choice, and focal γ. Cosine annealing emerges as the most impactful change (F1: $0.8932 \rightarrow 0.8984$, false negatives reduced 12.5%). An image-only ablation (Experiment 4) confirms metadata fusion adds +1.9 percentage points (pp) recall; SHAP analysis identifies anatomical site and age as dominant predictors. With six-transform test-time augmentation (TTA) the system achieves F1 = 0.9134, recall = 91.48%, specificity = 98.62%. EigenCAM validates ABCDE-aligned attention and a Mahalanobis out-of-distribution (OOD) detector flags non-dermoscopic inputs.**

***Index Terms*—melanoma, focal loss, metadata fusion, SHAP, test-time augmentation**

I. Introduction

Melanoma causes over 75% of skin cancer deaths despite representing fewer than 5% of cases [1]. Five-year survival exceeds 98% at stage I but falls below 25% for metastatic disease [2]. Dermoscopy improves sensitivity by 10–27 pp [3], yet inter-observer variability persists. Esteva et al. [4] showed a fine-tuned CNN matched 21 dermatologists, and the ISIC 2020 challenge [6] released structured metadata alongside dermoscopic images, motivating late-fusion approaches that incorporate patient context. A further challenge is severe class imbalance: at ∼13.6% positive rate, a trivial all-benign predictor achieves 86.4% accuracy [11].

We select **EfficientNet-B0** [8] as the backbone because (i) the ISIC 2020 winning solution used an EfficientNet ensemble [7], (ii) compound scaling offers a strong accuracy-per-FLOP trade-off at only 5.3 M parameters, and (iii) its smooth convergence makes it well-suited for controlled hyperparameter (HP) studies (Section IV-C).

Contributions:

1) Dual-branch fusion (CNN + metadata MLP) with focal loss α=0.864 derived from class proportions.
2) Five-experiment HP search (Baseline, Exp. 1–4) isolating scheduler, optimiser, focal γ, and metadata fusion, with per-experiment analysis of false-negative / false-positive trade-offs.
3) Measured ablation (+1.9 pp recall from metadata), per-feature SHAP analysis, error analysis, EigenCAM saliency, and Mahalanobis OOD.

Out of scope: Vision Transformers [17], [18] (require more data to compensate for lack of CNN inductive biases [17]), GAN augmentation, ensembles, multi-backbone comparison, multi-class diagnosis, and the ISIC private test set.

II. Literature Review

A. CNN Backbones for Dermoscopy

The ISIC 2020 challenge [6] released 33,126 dermoscopic images with patient metadata; our training set concatenates ISIC 2020 with additional ISIC 2019 images, yielding 37,648 samples. The winning solution used an EfficientNet ensemble [7] (AUC = 0.9490). **EfficientNet-B0** [8] applies compound scaling to balance accuracy and efficiency with only 5.3 M parameters. DenseNet-121 [9] and ResNet-50 [10] are popular alternatives but require more parameters for equivalent accuracy. ViTs [17], [18] show promise but need substantially more data due to lack of translation equivariance [17]; CNNs with ImageNet pretraining remain pragmatic at this scale.

B. Class Imbalance and Focal Loss

With 6.4 benign examples per malignant, standard cross-entropy steers gradients toward the majority class so the network can ignore rare positives and still minimise loss. Focal loss [11] multiplies the per-example loss by $(1 - p_t)^\gamma$: confident, correct predictions contribute nearly zero gradient; uncertain or wrong predictions receive amplified signal, forcing attention onto hard malignant cases. Setting α=1−0.136=0.864 further upweights malignant examples. Effect of varying γ is in Section IV-B.

C. Metadata Fusion, TTA, and Limitations

Pacheco et al. [12] curate a multimodal skin lesion dataset pairing clinical images with patient metadata, motivating fusion approaches. Groh et al. [13] caution about skin-tone bias. TTA [14] averages predictions over augmented views to reduce variance. Fujisawa et al. [15] identify amelanotic and small lesions as persistent DL failure modes. Davis and Goadrich [16] show AUC can overstate clinical utility on imbalanced data, motivating per-category error analysis (Section IV-F).

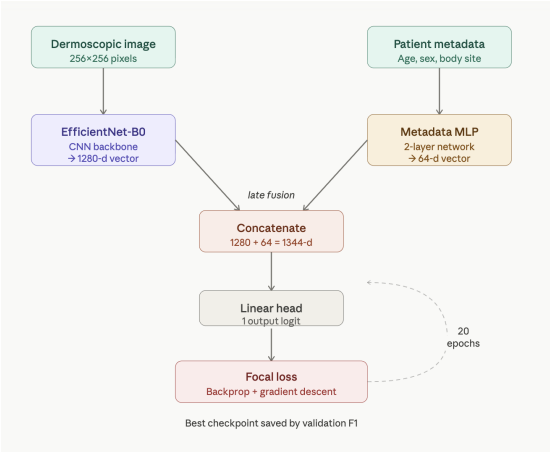

Fig. 1. Dual-branch pipeline: EfficientNet-B0 image branch and metadata MLP are fused via late concatenation, trained end-to-end with focal loss.

III. Methodology

A. Dataset

The SIIM-ISIC 2020 dataset contains ∼37 k dermoscopic images (256×256, **13.6% malignant**) with metadata: `age_approx` (continuous), `sex` (binary), and `anatom_site` (6 categories). An 80/20 stratified split (seed 42) preserves the class ratio. All experiments share this identical split.

B. Preprocessing and Augmentation

Images (train): Albumentations (CLAHE, blur, 90° snaps) then torchvision v2 (flip, affine ±15°, colour jitter, random erasing, ImageNet normalisation). **Images (val):** resize + normalise only. **Metadata:** median imputation + z-score for age; mode imputation + one-hot for categoricals → **14-d** vector.

C. Architecture

A dual-branch late-fusion model (Fig. 1): (1) an ImageNet-pretrained **EfficientNet-B0** backbone (→ 1280-d embedding) with the classification head removed; (2) a two-layer metadata MLP (14→128→64, BatchNorm + ReLU + Dropout 0.3). Late fusion is chosen because images and clinical scalars benefit from separate processing before being combined. The 1280-d and 64-d embeddings are concatenated (→ 1344-d) and projected to a single logit. Exp. 4 disables the metadata branch to isolate its contribution.

D. Loss and Training

Sigmoid focal loss [11]:

$$FL(p_t) = -\alpha_t \, (1 - p_t)^{\gamma} \log(p_t) \tag{1}$$

with α=0.864, γ=2.0. We define five experiments (Table I), each introducing *one primary change* so every factor can be isolated: **Baseline** (Adam, fixed LR, γ=2.0): minimal-tricks reference. **Exp. 1** adds cosine annealing to settle into sharper minima (higher F1, no new FP). **Exp. 2** Adam→AdamW + weight decay 10^{-5} to curb minority-class overfitting. **Exp. 3** lowers γ to 1.5, reducing suppression of easy negatives to recover specificity lost when Exp. 2 widened the decision boundary. **Exp. 4** disables metadata MLP (image-only ablation). The checkpoint with the highest validation F1 is retained.

TABLE I
Hyperparameter Configurations (EfficientNet-B0)

	Base	**Exp. 1**	**Exp. 2**	**Exp. 3**	**Exp. 4**
Optimiser	Adam	Adam	AdamW	AdamW	Adam
Scheduler	None	Cosine	Cosine	Cosine	None
Weight decay	0	0	10^{-5}	10^{-5}	0
Focal γ	2.0	2.0	2.0	1.5	2.0
Metadata	✓	✓	✓	✓	—
LR / Batch / Epochs	10^{-4} / 32 / 20				

E. Test-Time Augmentation

At inference, six geometric transforms (original, h-flip, v-flip, 90°/180°/270° rotation) are applied and their predictions averaged. Lesions are rotationally symmetric, so orientation should not affect confidence; averaging removes orientation-specific noise and smooths the decision boundary. Impact in Section IV-A.

F. Evaluation

Five metrics are reported on the validation split: **Recall** $= TP/(TP{+}FN)$: of all real cancers, how many did we catch. **Specificity** $= TN/(TN{+}FP)$: of all benign cases, how many were correctly cleared. **PPV** $= TP/(TP{+}FP)$: of everything flagged as cancer, how many actually were. **F1**: harmonic mean of PPV and recall—penalises models that sacrifice either. **Accuracy** is reported but not optimised: predicting benign always yields 86.4% accuracy yet catches zero cancers, making it clinically meaningless on this dataset. Recall is the primary metric: a missed melanoma carries substantially higher clinical cost than a false alarm.

IV. Experiments

All experiments use PyTorch 2.x with the EfficientNet-B0 backbone fine-tuned for up to 20 epochs. Identical data splits, augmentation pipelines, and evaluation code are shared across all configurations.

A. Incremental Ablation

Exp. 4 (image-only, same HP as Baseline) isolates the metadata branch. Its accuracy (97.28%) and raw F1 (0.9000) look strong, but the confusion matrix reveals the cost: 99 missed melanomas versus Baseline's 80—19 more patients undetected. At 13.6% prevalence, 19 extra misses barely register across 7,530 examples, illustrating why accuracy overstates clinical utility on imbalanced data [16]. Metadata adds +1.9 pp recall (FN: 99→80) at the cost of 39 extra FP, because the MLP responds to anatomical-site cues absent from image features alone. TTA then trades 7 extra FN for 55 fewer FP (recall 92.16%→91.48%, F1 = 0.9134, specificity 98.62%). Fig. 2 confirms the recall advantage from the earliest epochs.

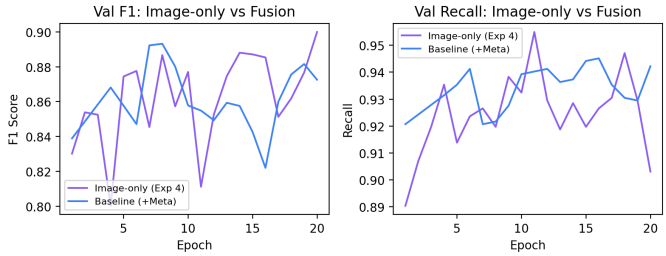

Fig. 2. Validation F1 and recall over 20 epochs: image-only (Exp. 4) vs. Baseline (with metadata MLP). Metadata fusion provides a consistent recall advantage throughout training.

TABLE II
HP SEARCH RESULTS (EFFICIENTNET-B0, BEST CHECKPOINT)

	F1	Recall	Acc	TP	FN	FP	TN
Baseline	0.8932	92.16%	97.01%	941	80	145	6364
Exp. 1	**0.8984**	93.14%	97.14%	951	70	145	6364
Exp. 2	0.8855	**93.54%**	96.72%	955	66	181	6328
Exp. 3	0.8968	92.75%	97.10%	947	74	144	6365
Exp. 4†	0.9000	90.30%	**97.28%**	922	99	106	6403

†Image-only (no metadata branch).

B. Hyperparameter Search

The five experiments form a deliberate chain (Table II). **Exp. 1:** does a smarter LR schedule help? With a fixed LR the optimiser overshoots the best loss valley; cosine annealing decays it to near-zero by epoch 20, letting the model settle. Payoff: FN drops 80→70 with *zero* extra FP—a "free lunch." **Exp. 2:** does weight decay push recall further? AdamW penalises large weights, broadening the decision boundary so borderline lesions get flagged. Recall peaks at 93.54% (FN = 66) but false alarms surge (FP: 145→181). **Exp. 3:** can the focal loss recover those FPs? Reducing γ from 2.0 to 1.5 gives easy benign examples more gradient influence, sharpening the boundary. FP falls to 144 (lowest of fusion models) while FN rises to 74.

Fig. 3 summarises all five experiments with the key design decision, plain-English rationale, and FN/FP outcome for each.

C. Training Dynamics

Fig. 4 overlays training dynamics. Baseline converges smoothly (loss 0.0152→0.0054, gap <0.32%). Cosine annealing (Exp. 1–3) drives loss lower but widens the gap (0.36–0.59%). Exp. 3 (γ=1.5) shows the largest gap. Exp. 1 achieves the best trade-off.

D. Discriminative Performance

All five experiments achieve AUC = 0.99, confirming that HP variations affect the precision–recall operating point rather than underlying ranking ability. This motivates confusion-matrix-level analysis (Table II) over AUC alone [16].

E. SHAP Feature Importance

SHAP KernelExplainer (Fig. 5) reveals consistent rankings: `anatom_site_torso` (dominant), `age_approx` (monotonic: older → higher risk), and `anterior_torso` (rare but high-impact, SHAP up to 0.18). Sex contributes modestly. Exp. 3 (γ=1.5) compresses magnitudes, distributing importance more evenly.

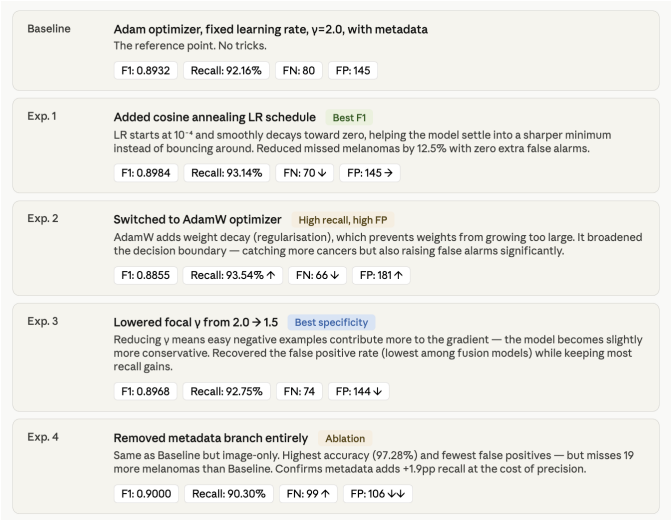

Fig. 3. Per-experiment summary: design decision, motivation, and best-checkpoint FN/FP outcome. Arrows indicate change relative to the prior experiment.

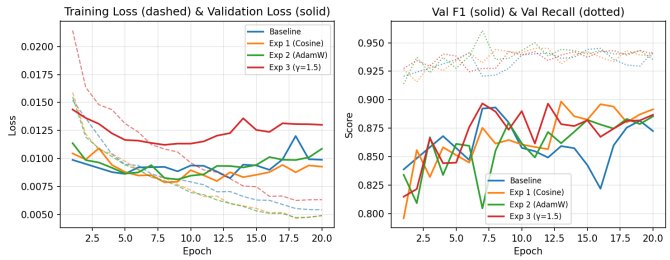

Fig. 4. Training curves for Baseline and Exp. 1–3. Top row: training and validation loss. Bottom row: validation F1 and recall. Cosine annealing (Exp. 1–3) produces lower training loss; Exp. 3 (γ=1.5) shows the largest train/val gap.

F. Error Analysis

FN (∼7–8.5% of malignant): amelanotic melanomas, small lesions (<6 mm), and early superficial spreading variants—subtypes lacking the pigmented features CNNs rely on [15]. **FP (∼1.4–2.8% of benign):** dysplastic naevi, atypical Spitz naevi, and irregular seborrhoeic keratoses [5]. Exp. 1 achieves a 6.86% FN rate, acceptable for a screening tool.

G. OOD Robustness

A Mahalanobis distance-based OOD detector [19] computes μ and Σ^{-1} of the backbone's 1280-d validation features; inputs whose feature vectors lie beyond $\mu+3\sigma$ (Table III) are flagged—a non-dermoscopic image (e.g. a car) sits far from the dermoscopy cluster and is correctly rejected. Exp. 1–3 produce tighter distributions, suggesting cosine annealing yields more compact feature representations.

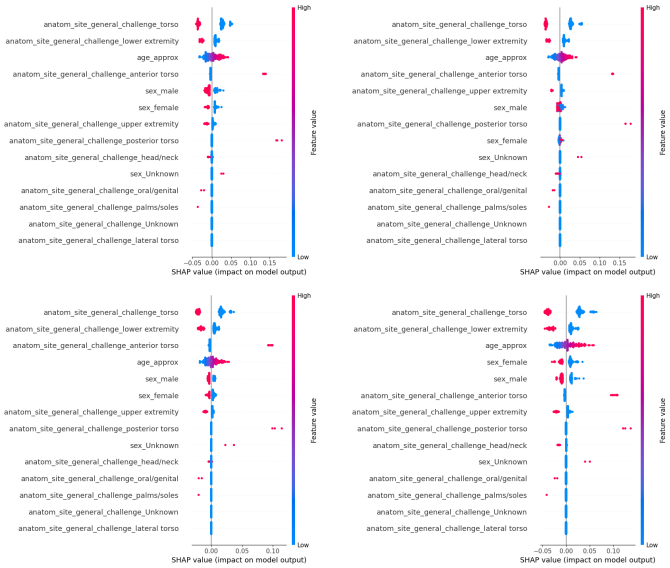

Fig. 5. SHAP beeswarm plots for Baseline and Exp. 1–3 (top-left to bottom-right). `anatom_site_torso` and `age_approx` are the dominant predictors across all configurations. Exp. 3 compresses SHAP magnitudes, distributing importance more evenly.

TABLE III
MAHALANOBIS OOD THRESHOLDS PER CONFIGURATION

Experiment	**Mean dist.** (μ)	**Threshold** ($\mu+3\sigma$)
Baseline	1277.4	3247.5
Exp. 1 – Cosine	1278.1	2652.3
Exp. 2 – AdamW	1276.8	2648.7
Exp. 3 – γ1.5	1275.3	2637.2

Exp. 4 omitted: image-only ablation, OOD stats not computed.

H. EigenCAM Interpretability

EigenCAM saliency maps confirm attention localises to asymmetric borders, colour variegation, and atypical pigment networks, aligning with clinical ABCDE criteria [4].

V. CONCLUSION AND FUTURE WORK

A dual-branch EfficientNet-B0 melanoma classifier fusing dermoscopic images with patient metadata achieves F1 = 0.8984, recall = 93.14% (Exp. 1, best epoch 13), reducing missed melanomas by 12.5% over Baseline with no additional FP. With TTA: F1 = 0.9134, recall = 91.48%, specificity = 98.62% (TTA trades 0.7 pp recall for 55 fewer FP). Ablation (Exp. 4) confirms metadata adds +1.9 pp recall; SHAP identifies `anatom_site_torso` and `age_approx` as dominant predictors across all HP experiments. EigenCAM validates ABCDE-aligned attention. A Mahalanobis OOD detector with empirically computed thresholds provides runtime safeguards against non-dermoscopic inputs.

Deployment. The full codebase is open-sourced at https://github.com/burrows99/melanoma-classification. Trained weights for all five experiments are hosted at https://huggingface.co/burrows99/melanoma-models, and a public Gradio demo is deployed at https://huggingface.co/spaces/burrows99/melanoma-classificiation, enabling browser-based inference with EigenCAM overlays and OOD detection.

Future work: (1) Larger EfficientNet variants (B3/B5) [7]; (2) multi-backbone comparison (DenseNet-121 [9], ResNet-50 [10]); (3) equity assessment via Fitzpatrick stratification [13]; (4) external validation on the ISIC private test set; (5) ViT/Swin backbones [17], [18] with larger pretraining budgets.

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